BUSINESS INSIGHTS THROUGH DATA



Presented By : The Mentalists - Group 4

INTRODUCTION

1- Data Maturity Assessment

2- Data Lifecycle Mapping

3- ESG Deep Dive

4- Recommendations & Roadmap

5- Conclusion & Presentation Integration





WHY TESCO?

- **UK’s largest retailer with operations in groceries**
- **Online retail and financial services**

Highly data-driven organisation:

- Clubcard loyalty programme
- E-commerce & customer analytics
- Global supply chain and logistics data

Strong ESG commitments and transparency:

- Net Zero 2050 target
- Detailed sustainability & governance reporting

- Science Based Targets. (2018, March). Tesco – Science Based Targets case study. https://files.sciencebasedtargets.org/production/legacy/2018/03/SBT_Tesco_CaseStudy.pd
- Tesco PLC. (2023). Annual report and financial statements 2023. https://dq06ugkuram52.cloudfront.net/files/19688091/48987055.pdf
- Tesco PLC. (n.d.). Climate change. https://www.tescoplc.com/sustainability/planet/climate-change
- Tesco PLC. (2024). Climate change factsheet 2024. https://www.tescoplc.com/climate-change-factsheet-2024
- Tesco PLC. (2025, May). Sustainability report 2025. https://www.tescoplc.com/media/wvkj1yic/tesco-sustainability-report-2025.pdf

DATA MATURITY

01

DEFINE

Dunnhumby provides deep insights from over *20 million Clubcard users* — powering customer segmentation and strategy.

02

INVEST

Digital platforms and AI tools for personalised recommendations and health nudges.

03

MANAGE

Handles massive customer information with strong privacy, governance, and cross-channel targeting.

04

MONITOR

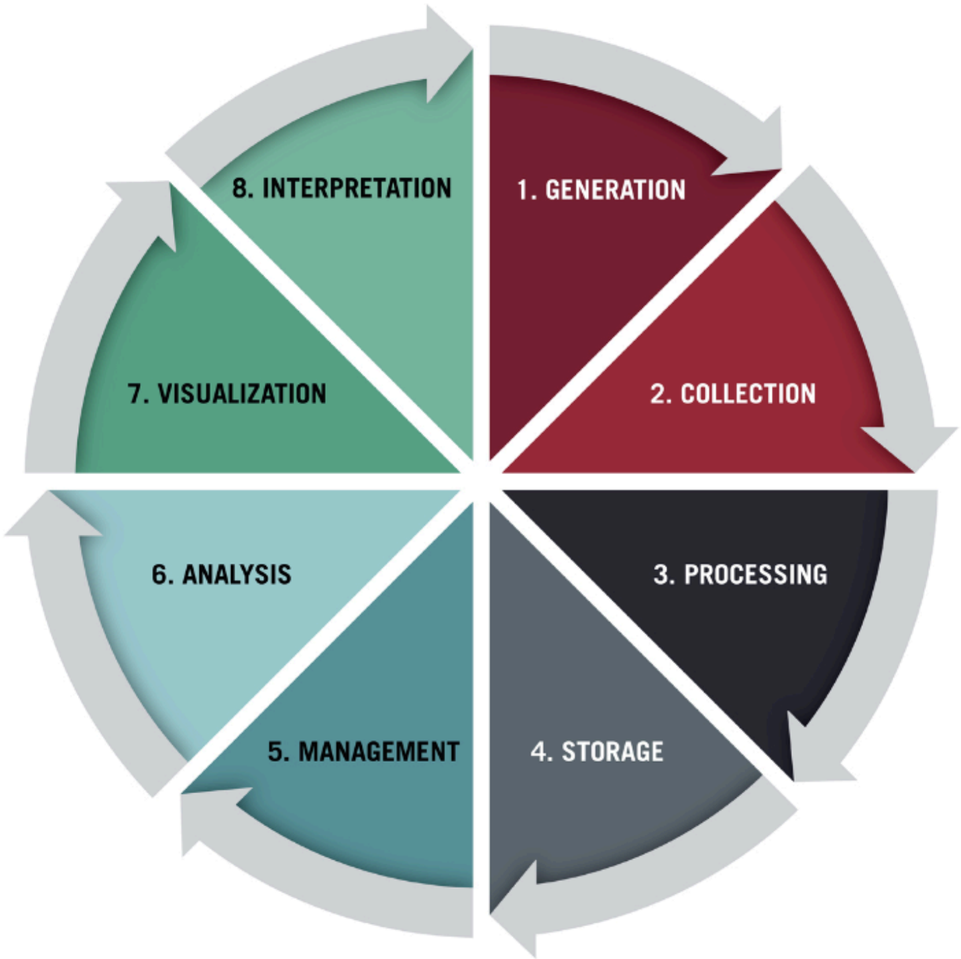
Tracking ROI, sales growth, Clubcard penetration, supplier campaign impact, and CSR results through advanced analytics and reporting.



- Anmut. (2024). *A comprehensive guide to data maturity.* Retrieved from https://www.anmut.co.uk/data-maturity-guide/
- Chartered Institute of Personnel and Development. (2024, July 9). Transforming the HR operating model: Tesco [Case study]. https://www.cipd.org/uk/knowledge/case-studies/operating-model-tesco/ cipd.org
- dunnhumby. (2022). Tesco Media Upfront 2022 [PDF]. https://www.dunnhumby.com/wp-content/uploads/2022/12/Tesco-Media-Upfront-2022.pdf
- Johnston, K. B. (2023, June 28). How many companies does Tesco own? Investopedia. https://www.investopedia.com/news/how-many-companies-does-tesco-own/
- AtliQ Technologies. (n.d.). How Tesco uses data to personalize the shopping experience: A deep dive.https://www.atliq.ai/how-tesco-uses-data-to-personalize-the-shopping-experience/



Data Lifecycle Steps



AtliQ AI. (2024, March 5). How Tesco uses data to personalize the shopping experience: A deep dive.https://www.atliq.ai/how-tesco-uses-data-to-personalize-the-shopping-experience/

Eagle Eye Solutions Group Plc. (2025). Case study: Tesco Clubcard Challenges — Personalization at scale.https://eagleeye.com/case-studies/tesco-clubcard-challenges

Skeldon, P. (2025, March 7). Tesco Media & Insight: The power of loyalty card data. InternetRetailing. https://internetretailing.net/tesco-media-insight-the-power-of-loyalty-card-data/

Tesco PLC. (2025). Serving our customers, communities and planet a little better every day: Tesco's economic and social impact 2025. https://www.tescoplc.com/media/vtbj2jpt/20250218-tesi-report.pdf

Tesco PLC. (2021, November 30). Tesco launches the UK's largest closed loop grocery media and insight platform, powered by dunnhumby. https://www.tescoplc.com/tesco-launches-the-uk-s-largest-closed-loop-grocery-media-and-insight-platform-powered-by-dunnhumby/

TESCO CLUBCARD DATA BREACH (2020): ESG ANALYSIS IN THE DATA STORAGE PHASE

- Tesco reported fraudulent activity affecting about 600,000 Clubcard accounts.
- Attackers used stolen credentials from other breaches to redeem loyalty points.
- No financial data was leaked, but the event revealed weaknesses in customer account protection and data ethics.

Ethics

Tesco emphasizes privacy and security, but reactive communication and user confusion followed the breach.

Sustainability

No direct environmental impact, but cybersecurity efficiency also affects digital energy use.

Governance

DPO and security teams responded promptly but lacked proactive monitoring.

- Tesco PLC. (n.d.). Privacy & cookies policy. https://www.tesco.com/groceries/en-GB/zone/privacy-and-cookies-policy Homepage
- Legal Expert. (n.d.). Tesco Clubcard data breach compensation claims. https://www.legalexpert.co.uk/data-breach-compensation/tesco-clubcard-data-breach-compensation-claims/
- Tesco PLC. (2023, July). Sustainability Committee Terms of Reference. https://www.tescoplc.com/media/eq3bsae4/sustainability-committee-terms-of-reference-final-july-2023.pdf tescoplc.com
- Tesco PLC. (2025, May). Sustainability Report 2024/25. https://www.tescoplc.com/media/wvkj1yic/tesco-sustainability-report-2025.pdf tescoplc.com
- Bîzgă, A. (2020, March 5). *Loyalty program fraud targets 600,000 Tesco shopper accounts.* HotForSecurity.https://www.bitdefender.com/en-us/blog/hotforsecurity/loyalty-program-fraud-targets-600000-tesco-shopper-accounts

TESCO BRAZILIAN MEAT (2020): ESG ANALYSIS IN THE DATA COLLECTION PHASE

- ✗ Tesco committed to end deforestation-linked sourcing but was found selling Brazilian meat from deforested regions.
- ✗ Weak supplier data validation and poor auditability led to accusations of greenwashing, exposing Tesco to regulatory and reputational risks.

Ethics	Sustainability	Governance
Tesco made sustainability claims without full data verification.	Relies on supplier self-reported data and limited audits, which failed to identify deforestation risks in Brazilian meat sourcing	Tesco has a Sustainability Committee, but it relies on supplier data and lacks strong checks to verify its accuracy.

- Tesco PLC. (n.d.). Privacy & cookies policy. https://www.tesco.com/groceries/en-GB/zone/privacy-and-cookies-policy Homepage
- Legal Expert. (n.d.). Tesco Clubcard data breach compensation claims. https://www.legalexpert.co.uk/data-breach-compensation/tesco-clubcard-data-breach-compensation-claims/
- Tesco PLC. (2023, July). Sustainability Committee Terms of Reference. https://www.tescoplc.com/media/eq3bsae4/sustainability-committee-terms-of-reference-final-july-2023.pdf tescoplc.com
- Tesco PLC. (2025, May). Sustainability Report 2024/25. https://www.tescoplc.com/media/wvkj1yic/tesco-sustainability-report-2025.pdf tescoplc.com
- Kakkad, R. (2023, September 19). Tesco accused of greenwashing deforestation pledges. Sustainability Beat. https://www.sustainability-beat.co.uk/2023/09/19/tesco-deforestation/ Sustainability Beat
- Mighty Earth. (2023, September 6). Call for Tesco to be investigated by the Competition and Markets Authority over sale of banned Brazilian meat and greenwashing claims. https://mightyearth.org/article/call-for-tesco-to-be-investigated-by-the-competition-and-markets-authority-over-sale-of-banned-brazilian-meat-and-greenwashing-claims/ Mighty Earth

DATA AND MANAGEMENT RECOMMENDATIONS

Ethics & Privacy

- Strengthn consent & Minimize stored data
- Target 60% data reduction (lower risk & emissions)

Governance

- Add AI anomaly detection & MFA
- Include bias audits for fair AI monitoring

Sustainability

- Streamline retention & Move to low-carbon cloud
- Up to 50% energy saving & improved brand trust

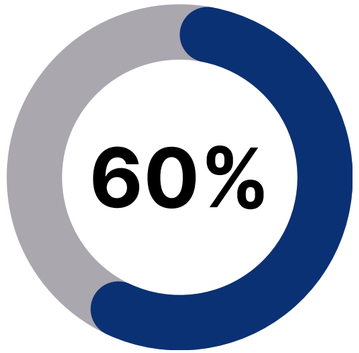
Supplier Accountability

- Require Tier-1 ESG data & renewable-energy reporting
- Prevent greenwashing in supply-chain claims

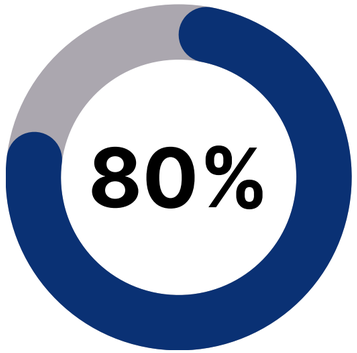
- Barocas, S., Hardt, M., & Narayanan, A. (2021). Fairness and machine learning. MIT Press.
- Carbon Accounting Software (SaaS). (2025). Retrieved from https://gaiacompany.io/carbon-accounting-software/?gad_source=1&gad_campaignid=21506580397&gbraid=0AAAAAqdN7DinbC4SJKxOiB1HG7rLw5RZF&gclid=CjOKCQjwgpzIBhCOARIsABZm7vGAUrtrBQLKJkyw5k8jOHpiNxVuIUUxsZDVQWISw7iJybQCiLjBJ8QaArMgEALw_wcB
- Gartner's Generative AI Report: Key trends and business impact. (n.d.-b). https://www.outsystems.com/1/gartner-report-generative-ai/
- ICO. (2023). Data minimisation under the UK GDPR. https://ico.org.uk
- Largest source of verified ESG data to explore sustainability practices. (n.d.). Retrieved from https://home.etoso.io/?utm_campaign=g&utm_source=google_us&utm_medium=cpc&utm_term=environmental+sustainability+reports&utm_content
- OECD. (2022). Guidelines on privacy and data flows. https://www.oecd.org
- Reducing cloud carbon footprint: Strategies for businesses. TechAhead. (2025, October 10). https://www.techaheadcorp.com/blog/strategies-for-reducing-enterprise-cloud-carbon-footprint/
- Sustainable supply chain management services | TÜV Süd. (n.d.-f). https://www.tuvsud.com/en/services/management-system-certification/sustainable-supply-chain-management
- Tesco PLC. (2024). The Little Helps Plan 2024. https://www.tescoplc.com/sustainability
- The UK GDPR. (n.d.). Retrieved from https://ico.org.uk/for-organisations/data-protection-and-the-eu/data-protection-and-the-eu-in-detail/the-uk-gdpr/
- UNESCO. (2021). Ethics of artificial intelligence. https://unesdoc.unesco.org
- WEF. (2023). SME ESG integration framework. https://www.weforum.org
- What is the Data Protection Act 2018 (DPA 2018)? (n.d.). Retrieved from https://www.itgovernance.co.uk/dpa-2018

KPIs

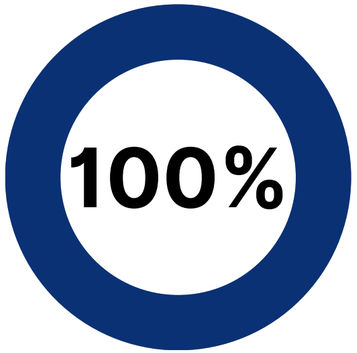
KPIs



Reducing stored personal data



Green compute by 2027



Verified Tier-1 suppliers

&

Cutting incident-response time to under 24 hours

Risk:

- Data gaps
- Vendor strain
- Phased rollout

Benefits:

- Compliance
- Strengthen trust
- Lower carbon cost

ROADMAP

Stabalise (months 0-3)

Dependacies: Existing IT & data infrastucture
Task Owners: DPO, IT Securities

Auromate (months 3-9)

Dependacies: AI & data governance tools
Task Owners: Head of Data Analytics, AI Ethics Officer

Optimise (months 9-18)

Dependacies: Cloud vendor integration, sustainability tracking
Task Owners: Sustainability Lead, Cloud Manager

Institutionalise (months 18>>)

Dependacies: Supplier data & ESG reporting systems
Task Owners: CSO, HR Training & Compliance

- Fidra, A. (2024, April 22). How UK companies are transforming through AI [Insight]. Retrieved from https://www.fidra.ai/insight/unleashing-the-potential-how-uk-companies-are-transforming-through-ai/
- InsurTech Digital. (2025, April 1). Tesco IMS selects Zensar for cloud transformation. Retrieved from https://insurtechdigital.com/articles/tesco-ims-selects-zensar-for-cloud-transformation
- Intelligent CISO. (2019, November 13). Tesco secures distributed workforce with OneLogin MFA solution. Retrieved from https://www.intelligentciso.com/2019/11/13/tesco-secures-distributed-workforce-with-onelogin-mfa-solution/
- Procurement Magazine. (2024, November 18). How Tesco enhanced supply chain visibility using AI. Retrieved from https://procurementmag.com/technology-and-ai/how-tesco-enhanced-supply-chain-visibility-using-ai
- Tesco PLC. (2024). Annual Report & Financial Statements 2024. Retrieved from https://data.fca.org.uk/artefacts/NSM/Portal/NI-000096769/NI-000096769.pdf
- Tesco PLC. (2021, May 18). Cyber Security – protecting customer and colleague data. Retrieved from https://www.tescoplc.com/sustainability/documents/policies/cyber-security-protecting-customer-and-colleague-data
- Tesco PLC. (2025, May 25). Sustainability Report 2024/25 – Making a positive impact. Retrieved from https://www.tescoplc.com/media/wvkj1yic/tesco-sustainability-report-2025.pdf
- Tesco PLC. (2025, May 16). Tesco Annual Report 2025. Retrieved from https://www.tescoplc.com/media/ky0bfwpo/tesco_ar25_interactive.pdf

CONCLUSION

- Strong data maturity gives Tesco an edge in customer insight, efficiency, and sustainability
- Key challenges: AI fairness, supplier data verification, and cloud storage impact
- Needs: stronger ethics, transparency, and sustainable data management
- Recent concerns: Clubcard data breach and greenwashing claims
- Solutions: better data protection, greener cloud systems, verified supplier ESG reporting
- Goal: rebuild trust, cut emissions, and lead with integrity and innovation

THANK YOU!

REFERENCES

- Anmut. (2024). A comprehensive guide to data maturity. Retrieved from https://www.anmut.co.uk/data-maturity-guide/
- AtliQ Technologies. (n.d.). How Tesco uses data to personalize the shopping experience: A deep dive.https://www.atliq.ai/how-tesco-uses-data-to-personalize-the-shopping-experience/
- Barocas, S., Hardt, M., & Narayanan, A. (2021). Fairness and machine learning. MIT Press. Retrieved from https://www.myecole.it/biblio/wp-content/uploads/2020/11/2020-Fairness-book.pdf
- Bîzgă, A. (2020, March 5). Loyalty program fraud targets 600,000 Tesco shopper accounts. HotForSecurity. Retrieved from https://www.bitdefender.com/en-us/blog/hotforsecurity/loyalty-program-fraud-targets-600000-tesco-shopper-accounts
- Carbon Accounting Software (SaaS). (2025). Retrieved from https://gaiacompany.io/carbon-accounting-software/?gad_source=1&gad_campaignid=21506580397&gbraid=0AAAAAqdN7DinbC4SJKxOiB1HG7rLw5RZF&gclid=Cj0KCQjwgpzIBhCOARIsABZm7vGAUrtrBQLKJkyw5k8jOHpiNxVuIUUxsZDVQWISw7iJybQCiLjBJ8QaArMgEALw_wcB
- Chartered Institute of Personnel and Development. (2024, July 9). *Transforming the HR operating model: Tesco* [Case study]. https://www.cipd.org/uk/knowledge/case-studies/operating-model-tesco/ cipd.org
- Climate change factsheet 2024, Tesco plc : https://www.tescoplc.com/climate-change-factsheet-2024
- tescoplc.com
- Climate change, Tesco plc : https://www.tescoplc.com/sustainability/planet/climate-change/tescoplc.com
- Decision Marketing. (2024, May 17). Clubcard data in the driving seat as Tesco profits surge. Retrieved from https://www.decisionmarketing.co.uk/news/clubcard-data-in-the-driving-seat-as-tesco-profits-surge
- Eagle Eye. (2024). Tesco Clubcard Challenges case study. Retrieved from https://www.eagleeye.com/case-studies/tesco-clubcard-challenges
- Fidra, A. (2024, April 22). How UK companies are transforming through AI [Insight]. Retrieved from https://www.fidra.ai/insight/unleashing-the-potential-how-uk-companies-are-transforming-through-ai/
- Gartner's Generative AI Report: Key trends and business impact. (n.d.). Retrieved from https://www.outsystems.com/1/gartner-report-generative-ai/
- ICO. (2023). Data minimisation under the UK GDPR. Retrieved from https://ico.org.uk
- InsurTech Digital. (2025, April 1). Tesco IMS selects Zensar for cloud transformation. Retrieved from https://insurtechdigital.com/articles/tesco-ims-selects-zensar-for-cloud-transformation
- Intelligent CISO. (2019, November 13). Tesco secures distributed workforce with OneLogin MFA solution. Retrieved from https://www.intelligentciso.com/2019/11/13/tesco-secures-distributed-workforce-with-onelogin-mfa-solution/
- InternetRetailing. (2024, March 21). Tesco Media & Insight: The power of loyalty card data. Retrieved from https://internetretailing.net/tesco-media-insight-the-power-of-loyalty-card-data/
- Johnston, K. B. (2023, June 28). How many companies does Tesco own? Investopedia. https://www.investopedia.com/news/how-many-companies-does-tesco-own/
- Kakkad, R. (2023, September 19). Tesco accused of greenwashing deforestation pledges. Sustainability Beat.Retrieved from https://www.sustainability-beat.co.uk/2023/09/19/tesco-deforestation/SustainabilityBeat
- Largest source of verified ESG data to explore sustainability practices. (n.d.). Retrieved from https://home.etoso.io/?utm_campaign=g&utm_source=google_us&utm_medium=cpc&utm_term=environmental+sustainability+reports&utm_content
- Legal Expert. (n.d.). Tesco Clubcard data breach compensation claims. Retrieved from https://www.legalexpert.co.uk/data-breach-compensation/tesco-clubcard-data-breach-compensation-claims/ legalexpert.co.uk
- OECD. (2022). Guidelines on privacy and data flows. Retrieved from https://www.oecd.org
- Procurement Magazine. (2024, November 18). How Tesco enhanced supply chain visibility using AI. Retrieved from https://procurementmag.com/technology-and-ai/how-tesco-enhanced-supply-chain-visibility-using-ai

REFERENCES

- Reducing cloud carbon footprint: Strategies for businesses. TechAhead. (2025, October 10). Retrieved from https://www.techaheadcorp.com/blog/strategies-for-reducing-enterprise-cloud-carbon-footprint/
- Reuters. (2024, July 26). UK watchdog sees little evidence loyalty prices mislead shoppers. Retrieved from https://www.reuters.com/world/uk/uk-watchdog-little-evidence-supermarkets-loyalty-prices-misleading-shoppers-2024-07-26/
- Science Based Targets. (2018, March). Tesco – Science Based Targets case study. Science Based Targets. (2018, March). Tesco – Science Based Targets case study.
- Sustainable supply chain management services | TÜV Süd. (n.d.). Retrieved from https://www.tuvsud.com/en/services/management-system-certification/sustainable-supply-chain-management
- Tesco Annual Report & Financial Statements 2023 (PDF) : https://dq06ugkuram52.cloudfront.net/files/19688091/48987055.pdf
- Tesco PLC. (2024). Annual Report & Financial Statements 2024. Retrieved from https://data.fca.org.uk/artefacts/NSM/Portal/NI-000096769/NI-000096769.pdf
- Tesco PLC. (n.d.). Climate change. https://www.tescoplc.com/sustainability/planet/climate-change/
- Tesco PLC. (2024). Climate change factsheet 2024. https://www.tescoplc.com/climate-change-factsheet-2024
- Tesco PLC. (2021, May 18). Cyber Security – protecting customer and colleague data. Retrieved from https://www.tescoplc.com/sustainability/documents/policies/cyber-security-protecting-customer-and-colleague-data
- Tesco PLC. (2025, February 18). Our customers, communities and planet a little better every day: Tesco's economic and social impact 2025 [PDF]. Retrieved from https://www.tescoplc.com/media/vtbj2jpt/20250218-tesi-report.pdf
- Tesco PLC. (2025, June 5). Privacy & cookies policy. Retrieved from https://www.tesco.com/groceries/en-GB/zone/privacy-and-cookies-policy Homepage
- Tesco PLC. (2023, July). Sustainability Committee Terms of Reference. Retrieved from https://www.tescoplc.com/media/eq3bsae4/sustainability-committee-terms-of-reference-final-july-2023.pdf tescoplc.com
- Tesco PLC. (2025, May 25). Sustainability Report 2024/25 – Making a positive impact. Retrieved from https://www.tescoplc.com/media/wvkj1yic/tesco-sustainability-report-2025.pdf
- Tesco PLC. (2021, November 30). Tesco launches the UK's largest closed-loop grocery media and insight platform, powered by dunnhumby. Retrieved from https://www.tescoplc.com/tesco-launches-the-uk-s-largest-closed-loop-grocery-media-and-insight-platform-powered-by-dunnhumby/
- Tesco PLC. (2024). The Little Helps Plan 2024. Retrieved from https://www.tescoplc.com/sustainability
- Tesco PLC. (2025). Tesco Sustainability Report 2024/25. Retrieved from https://www.tescoplc.com/media/wvkj1yic/tesco-sustainability-report-2025.pdf
- Tesco PLC. (2025, May 16). *Tesco Annual Report 2025*. Retrieved from https://www.tescoplc.com/media/ky0bfwpo/tesco_ar25_interactive.pdf
- Tesco – Science Based Targets case study : https://files.sciencebasedtargets.org/production/legacy/2018/03/SBT_Tesco_CaseStudy.pdf
- Tesco Sustainability Report 2025 PDF : https://www.tescoplc.com/media/wvkj1yic/tesco-sustainability-report-2025.pdf
- tescoplc.com
- The UK GDPR. (n.d.). Retrieved from https://ico.org.uk/for-organisations/data-protection-and-the-eu/data-protection-and-the-eu-in-detail/the-uk-gdpr/
- UNESCO. (2021). Ethics of artificial intelligence. Retrieved from https://unesdoc.unesco.org
- WEF. (2023). SME ESG integration framework. Retrieved from https://www.weforum.org
- What is the Data Protection Act 2018 (DPA 2018)? (n.d.). Retrieved from https://www.itgovernance.co.uk/dpa-2018